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RESPIRATORY APPARATUS AIRFLOW TUBE

Abstract

A respiratory apparatus airflow tube configured to deliver an airflow from an airflow generator machine to a patient interface includes a flexible tube, a patient interface connector and a machine interface connector. The flexible tube includes a machine end, a patient end, and first and second heater wires extending from the machine end toward the patient end. The patient interface connector is attached to the patient end and is configured to attach to a patient interface. The machine connector is attached to the machine end and is configured to attach to an airflow generator machine. The machine connector includes an airflow housing having an interior wall defining an airflow pathway through which an airflow may be delivered into the machine end of the flexible tube and a temperature sensor located outside the airflow pathway.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application is based on and claims the benefit of U.S. provisional patent application Ser. No. 63/560,311, filed Mar. 1, 2024, and U.S. provisional patent application Ser. No. 63/734,927, filed Dec. 17, 2024, the contents of which are hereby incorporated by reference in their entirety.

FIELD

[0002] Embodiments of the present disclosure relate to respiratory apparatuses that deliver an airflow to a patient through a patient interface, such as a mask, and more particularly to heatable airflow tubes through which the airflow is delivered to the patient.

BACKGROUND

[0003] Respiratory apparatuses are used to deliver breathable gas to a patient through a patient interface, such as a breathing mask. One such respiratory apparatus is a Continuous Positive Air Pressure (CPAP) unit, which may be used to provide pressure support therapy to a patient suffering from disordered breathing (e.g., sleep apnea, snoring, etc.) during sleep, in which a continuous positive air pressure is applied to the patient's airway. This positive air pressure effectively “splints” the airway, thereby maintaining an open passage to the lungs.

[0004] FIG. 11 is a simplified diagram of an example of a respiratory apparatus 200, such as a CPAP unit, in accordance with the prior art. The respiratory apparatus 200 includes an airflow generator machine 202 having an airflow generator 203 that generates an airflow 204 for a pressure support therapy that is delivered to a patient 206 through an airflow tube 208. The airflow tube 208 includes a flexible tube 210 having a machine interface or a machine connector 212 at a machine end 214, and a patient interface connector 216 at a patient interface end 218. The machine connector 212 connects to the airflow generator machine 202 and the patient interface connector 216 connects to a patient interface 220 (e.g., a mask). The airflow generator machine 202 may include a humidifier that humidifies the airflow 204 in order to reduce drying of the patient's airway and to improve patient comfort.

[0005] FIGS. 12 and 13 are simplified diagrams of example airflow tubes 208 of the prior art that include heater wires 222 and 224 extending in a spiral or helical manner within a wall of the flexible tube 210 from the machine end 214 to the patient interface end 218. A heater 226 (FIG. 11) of the respiratory apparatus 200 controls a heating current $I_{sub.h}$ through the heater wires 222 and 224 to heat the flexible tube 210 and the airflow 204 traveling through the flexible tube 210 to improve patient comfort. Additionally, the heater 226 may heat the flexible tube 210 to prevent a “rain-out” condition, which occurs when moisture in the airflow 204 condenses on the inside of the flexible tube 210, such as when the flexible tube 210 is placed in a relatively cool environment.

[0006] Respiratory apparatuses utilizing the heatable airflow tube 208 conventionally include a temperature sensor 230 (e.g., thermistor) at the patient interface connector 216 that is used to sense the temperature of the airflow 204 prior to the patient interface 220. The heater 226 drives the heater current $I_{sub.h}$ through the heater wires 222 and 224 based on a temperature signal from the temperature sensor 230 to maintain a desired airflow temperature (e.g., around 82-84° Fahrenheit) at the patient interface end 218 and to prevent a rain-out condition in the flexible tube 210.

[0007] The temperature signal is communicated to the heater 226 through at least one sensing wire,

such as sensing wire **232** in the example airflow tube **208** shown in FIG. **12**, or sensing wires **234** and **236** in the example airflow tube **208** shown in FIG. **13**. The one or more sensing wires are typically integrated into the flexible tube **210** along with the heater wires **222** and **224**. The temperature signal generally takes the form of a voltage (Vs) across the temperature sensor **230**, such as a voltage between the sensing wire **232** and one of the heating wires (e.g., heating wire **224**), as indicated in FIG. **12**, or a voltage between the sensing wires **234** and **236**, as indicated in FIG. **13**.

[0008] Accordingly, the airflow tube **208** of the prior art typically includes three or more wires: two heater wires **222** and **224** and at least one sensing wire (**232** or **234** and **236**) that is connected to the temperature sensor **230** at the patient interface connector **216** or the patient interface end **218** of the airflow tube **208**, as shown in FIGS. **11-13**. The multiple wires in the airflow tube **208** increase the cost and complexity of the airflow tube **208** while decreasing the flexibility of the flexible tube **210**. Examples of such airflow tubes **208** and corresponding respiratory apparatuses are described in U.S. Pat. No. 8,733,349 and U.S. Patent Application Publication No. US2021/0379319A1.

[0009] Furthermore, conventional means to control tubing temperature where the temperature is detected by the temperature sensor **230** at the patient interface connector **216** or the patient interface end **218** is significantly affected by the temperature of the airflow **204** but may also be affected by the exhalation breath of the patient. As a result, the temperature detected by the temperature sensor **230** may fluctuate in response to the patient's breathing, which may destabilize the control of the airflow tube heating.

[0010] Conventional airflow tubes also utilize patient interface connectors **216** that support the temperature sensor **230** within a structure **238** that protrudes into the airflow pathway **240** through which the airflow **204** travels, as shown FIG. **14**, which is a simplified end view of the patient interface connector **216** of the airflow tube **208**. The placement of the structure **238** in the airflow pathway **240** produces a pressure drop that reduces the pressure at the patient interface **220**, increases the complexity of the patient interface connector **216** and increases manufacturing costs. As a result of the pressure drop associated with the protruding structure **238**, the efficiency at which the airflow generator machine **202** provides a desired pressurized airflow **204** at the patient interface **220** is reduced resulting in higher power consumption.

SUMMARY

[0011] Embodiments of the present disclosure are directed to respiratory apparatus airflow tubes that are configured to deliver an airflow from an airflow generator machine to a patient interface and respiratory apparatuses that include the airflow tube. One embodiment of the airflow tube includes a flexible tube, a patient interface connector and a machine interface connector. The flexible tube includes a machine end, a patient end, and first and second heater wires extending from the machine end toward the patient end. The patient interface connector is attached to the patient end and is configured to attach to a patient interface. The machine connector is attached to the machine end and is configured to attach to an airflow generator machine. The machine connector includes an airflow housing having an interior wall defining an airflow pathway through which an airflow may be delivered into the machine end of the flexible tube and a temperature sensor located outside the airflow pathway.

[0012] In one embodiment, the machine connector includes an airflow tube electrical connector configured to connect to a machine electrical connector of the airflow generator machine. The airflow tube electrical connector includes a first electrical contact connected to the first heater wire, a second electrical contact connected to the second heater wire and a third electrical contact connected to the temperature sensor.

[0013] In one embodiment, the temperature sensor is connected in parallel with the second and third electrical contacts.

[0014] In one embodiment, the airflow tube electrical connector includes a fourth electrical contact connected to the temperature sensor, and the temperature sensor is connected in series with the

third and fourth electrical contacts.

[0015] In one embodiment, the temperature sensor is located on an exterior side of the airflow housing.

[0016] In one embodiment, the machine connector includes an airflow housing having an interior wall defining an airflow pathway through which an airflow may be delivered into the machine end of the flexible tube, and the temperature sensor is supported in the airflow pathway.

[0017] In one embodiment, the airflow tube includes an assembly that is attached to the airflow housing including the temperature sensor, a first conductor connected to the first heater wire, a second conductor connected to the second heater wire, a third conductor connected to the temperature sensor, and an airflow tube electrical connector. The airflow tube electrical connector is configured to connect to a machine electrical connector of the airflow generator machine and includes a first electrical contact connected to the first conductor, a second electrical contact connected to the second conductor, and a third electrical contact connected to the third conductor.

[0018] In one embodiment, the airflow housing includes an assembly connector that secures the electrical connector to the airflow housing.

[0019] One embodiment of the respiratory apparatus includes an airflow machine and an airflow tube. The airflow machine includes an airflow generator configured to drive an airflow through an output port, a heating unit and a heater controller. The airflow tube includes a flexible tube having a machine end, a patient end, and first and second heater wires extending from the machine end toward the patient end, a patient interface connector attached to the patient end and configured to attach to a patient interface, and a machine connector attached to the machine end of the flexible tube and connected to the output port. The machine connector includes an airflow housing having an interior wall defining an airflow pathway through which an airflow may be delivered into the machine end of the flexible tube and a temperature sensor located outside the airflow pathway. The heater controller is configured to control the heating unit to drive a current through the first and second heater wires using the temperature sensor.

[0020] In one embodiment, the airflow machine includes a machine electrical connector and the airflow tube includes an airflow tube electrical connector configured to connect to the machine electrical connector. The airflow tube electrical connector includes a first electrical contact connected to the first heater wire, a second electrical contact connected to the second heater wire, and a third electrical contact connected to the temperature sensor.

[0021] In one embodiment, the temperature sensor is connected across or in parallel with the second and third electrical contacts.

[0022] In one embodiment, the heater controller controls the current through the first and second heater wires based on a voltage between the second and third electrical contacts, which varies based on a temperature of the temperature sensor.

[0023] In one embodiment, the airflow tube electrical connector includes a fourth electrical contact connected to the temperature sensor, the temperature sensor is connected in series with the third and fourth electrical contacts, and the heater controller controls the current through the first and second heater wires based on a voltage between the third and fourth electrical contacts, which varies based on a temperature of the temperature sensor.

[0024] In one embodiment, the temperature sensor is located on an exterior side of the airflow housing.

[0025] In one embodiment, the machine connector includes an airflow housing having an interior wall defining an airflow pathway through which the airflow is delivered into the machine end of the flexible tube, and the temperature sensor is supported in the airflow pathway.

[0026] In one embodiment, the airflow tube includes an assembly attached to the airflow housing. The assembly including the temperature sensor, a first conductor connected to the first heater wire, a second conductor connected to the second heater wire, a third conductor connected to the temperature sensor, and an airflow tube electrical connector connected to the machine electrical

connector. The airflow tube electrical connector includes a first electrical contact connected to the first conductor, a second electrical contact connected to the second conductor, and a third electrical contact connected to the third conductor.

[0027] In one embodiment, the airflow housing includes an assembly connector that secures the electrical connector to the airflow housing.

[0028] In one embodiment, the respiratory apparatus includes a patient interface comprising a mask attached to the patient interface connector.

[0029] Another embodiment of the respiratory apparatus airflow tube includes a flexible tube including a machine end, a patient end, and first and second heater wires extending from the machine end toward the patient end, a patient interface connector attached to the patient end and configured to attach to a patient interface, and a machine connector attached to the machine end and configured to attach to an airflow generator machine. The machine connector includes a temperature sensor and an airflow tube electrical connector configured to connect to a machine electrical connector of the airflow generator machine. The airflow tube electrical connector including a first electrical contact connected to the first heater wire, a second electrical contact connected to the second heater wire, and a third electrical contact connected to the temperature sensor.

[0030] In one embodiment, the temperature sensor is connected in parallel with the second and third electrical contacts.

[0031] In one embodiment, the airflow tube electrical connector comprises a fourth electrical contact connected to the temperature sensor, and the temperature sensor is connected in series with the third and fourth electrical contacts.

[0032] In one embodiment, the machine connector includes an airflow housing having an interior wall defining an airflow pathway through which an airflow may be delivered into the machine end of the flexible tube, and the temperature sensor is located outside the airflow pathway.

[0033] In one embodiment, the machine connector includes an airflow housing having an interior wall defining an airflow pathway through which an airflow may be delivered into the machine end of the flexible tube, and the temperature sensor is supported in the airflow pathway.

[0034] Another embodiment of a respiratory apparatus airflow tube includes a flexible tube including a machine end, a patient end, and first and second heater wires extending from the machine end toward the patient end, a patient interface connector attached to the patient end and configured to attach to a patient interface, a machine connector attached to the machine end and configured to attach to an airflow generator machine, and a temperature sensor located along an airflow pathway extending through the machine connector, the flexible tube and the patient interface at a location that is closer to the machine connector than the patient interface connector.

[0035] In one embodiment, the temperature sensor is supported in the airflow pathway.

[0036] In one embodiment, the temperature sensor is supported outside the airflow pathway.

[0037] In one embodiment, the machine connector includes the temperature sensor.

[0038] In one embodiment, the flexible tube includes the temperature sensor.

[0039] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in the Background.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0040] FIG. 1 is a simplified diagram of an example of a respiratory apparatus, in accordance with

embodiments of the present disclosure.

[0041] FIG. 2 is a side view of an example of an airflow tube, in accordance with embodiments of the present disclosure.

[0042] FIGS. 3, 4 and 5 respectively are isometric, top and front views of an example of a machine connector of an airflow tube, in accordance with embodiments of the present disclosure.

[0043] FIGS. 6 and 7 are simplified diagrams of example airflow tubes, in accordance with embodiments of the present disclosure.

[0044] FIG. 8 is a simplified side view of a respiratory apparatus, in accordance with embodiments of the present disclosure.

[0045] FIG. 9 is a flowchart of an example of a method of controlling airflow tube heating, in accordance with embodiments of the present disclosure.

[0046] FIG. 10 is a simplified diagram of an example of a controller, in accordance with embodiments of the present disclosure.

[0047] FIG. 11 is a simplified diagram of an example of a respiratory apparatus, in accordance with the prior art.

[0048] FIGS. 12 and 13 are simplified diagrams of example airflow tubes, in accordance with the prior art.

[0049] FIG. 14 is a simplified end view of a patient interface connector of an airflow tube, in accordance with the prior art.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0050] Embodiments of the present disclosure are described more fully hereinafter with reference to the accompanying drawings. Elements that are identified using the same or similar reference characters refer to the same or similar elements. The various embodiments of the present disclosure may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the present disclosure to those skilled in the art.

[0051] It will be understood that when an element is referred to as being “connected,” “coupled,” or “attached” to another element, it can be directly connected, coupled or attached to the other element, or it can be indirectly connected, coupled, or attached to the other element where intervening or intermediate elements may be present. In contrast, if an element is referred to as being “directly connected,” “directly coupled” or “directly attached” to another element, there are no intervening elements present. Drawings illustrating direct connections, couplings or attachments between elements also include embodiments, in which the elements are indirectly connected, coupled or attached to each other.

[0052] FIG. 1 is a simplified diagram of an example of a respiratory apparatus **100**, in accordance with embodiments of the present disclosure. In one example, the respiratory apparatus **100** is in the form of a Continuous Positive Air Pressure (CPAP) unit. Embodiments of the present disclosure may be applicable to other types of respiratory apparatuses, such as a ventilator, for example.

[0053] The respiratory apparatus **100** generally includes an airflow generator machine **102**, an airflow tube **104**, and a patient interface **106**. The airflow generator machine **102** may be a conventional device that is configured to generate an airflow **110**, such as an airflow for providing pressure support therapy for a patient, for example.

[0054] The airflow tube **104** includes a flexible tube **112**, a machine connector **114** attached to a machine end **116** of the flexible tube **112** and a patient interface connector **118** attached to a patient interface end **120** of the flexible tube **112**. In some embodiments, a pair of conventional resistive heater wires **122** and **124** (hereinafter “heater wires”) extend through the flexible tube **112**, such as in a spiraling or helical manner within a wall of the flexible tube **112**. The machine connector **114** is configured to connect to an output port **126** of the airflow generator machine **102**, through which the airflow **110** is discharged. The patient interface connector **118** is configured to attach to the

patient interface **106**, which may take on any suitable form, such as a full-face mask, nasal mask, oral-nasal mask, mouth mask, nasal prongs, cannula etc. Also, headgear may be utilized to comfortably support the patient interface end **120** and the patient interface **106** in a desired position on a patient **127**. The airflow **110** discharged through the output port **126** is delivered to the patient interface **106** through an airflow pathway **128** that extends through the machine connector **114**, the flexible tube **112** and the patient interface connector **118**.

[0055] The airflow generator machine **102** delivers a heater current through the heater wires **122** and **124**, such as a pulse width modulated (PWM) current, based on a temperature signal from a temperature sensor **130**, which may take on any suitable form. In one example, the temperature sensor **130** comprises a thermistor (e.g., 5 Kohm or 10 Kohm thermistor), such as a negative temperature coefficient (NTC) thermistor, for example. Other examples of the temperature sensor **130** include a resistance temperature detector (RTD), a thermocouple, or another suitable temperature sensor.

[0056] The airflow tube **104** includes a machine side **132** and a patient interface side **134** of a mid-point along the airflow pathway **128**. Thus, elements on the machine side **132** are located closer to the machine connector **114** and the airflow generator machine **102** than elements on the patient interface side **134**, and elements located on the patient interface side **134** are located closer to the patient interface connector **118** and the patient interface **106** than elements on the machine side **132**.

[0057] In one embodiment, the temperature sensor **130** is located proximate to the airflow generator machine **102** on the machine side **132** of the airflow tube **104**, such as within a distance **136** of the terminating end **138** of the machine connector **114**. Examples of the distance **136** include less than about 12 inches, less than about 8 inches, less than about 4 inches or less than about 2 inches from the terminating end **138**.

[0058] In one embodiment, the flexible tube **112** includes or supports the temperature sensor **130**, as indicated by temperature sensor **130A** at a location that is on the machine side **132**. Here, the temperature sensor **130A** may be located outside the airflow pathway **128**, such as on an exterior side of the flexible tube **112** or within a wall of the flexible tube **112**, or within the airflow pathway **128**, such as attached to an interior wall of the flexible tube **112** that defines a portion of the airflow pathway **128**, for example.

[0059] In some embodiments, the machine connector **114** includes the temperature sensor **130**, as indicated by the temperature sensor **130B**. In one embodiment, the temperature sensor **130B** is located outside the airflow pathway **128**. In another embodiment, the temperature sensor **130B** is located within the airflow pathway **128**.

[0060] FIG. **2** is a side view of an example of the airflow tube **104**, in accordance with embodiments of the present disclosure. FIGS. **3**, **4** and **5** are isometric, top and front views of an example of the machine connector **114**, in accordance with embodiments of the present disclosure. The flexible tube may include a wall having one or more helical ribs **137**, and the heater wires may be contained within the one or more helical ribs **137**.

[0061] The patient interface connector **118** may comprise a conventional socket for connecting to a port or tube of the patient interface **106**. The removal of the temperature sensor from the patient interface end **120** of the airflow tube **104** allows the patient interface end **120** and the patient interface connector **118** to be simplified relative to that of the prior art airflow tube **208**, such as by eliminating the temperature sensor supporting structure **238** (FIG. **14**) that protrudes into the airflow pathway **240** and produces a pressure drop that inhibits the airflow **204**.

[0062] The machine connector **114** may include an airflow housing **140** that is connected to the machine end **116** of the flexible tube **112**. The airflow housing **140** may be formed of rigid plastic, flexible polymers (e.g., thermoplastic elastomers), combinations of these materials or other conventional materials. The airflow housing **140** includes an interior wall **142** (FIG. **5**) that defines the airflow pathway **128** through the machine connector **114**. Conventional port connecting

members **144**, such as tabs that project from the terminating end **138** of the airflow housing **140** may cooperate with corresponding connecting members at the output port **126** of the airflow generator machine **102** to secure the machine connector **114** to the output port **126** and couple the airflow **110** discharged through the output port **126** to the airflow pathway **128** of the machine connector **114** and the flexible tube **112**. A cover **146** may extend over the airflow housing **140** and the temperature sensor **130B** and coupled to the flexible tube **112**, as shown in FIG. 2.

[0063] Embodiments of the machine connector **114** that include the temperature sensor **130B** either support the temperature sensor **130B** outside the airflow pathway **128** or within the airflow pathway **128** through attachment of the temperature sensor **130B** to the airflow housing **140**. In one example, the temperature sensor **130B** is located outside the airflow pathway **128** by attaching or supporting the temperature sensor **130B** on an exterior **148** of the airflow housing **140**, as shown in FIGS. 3 and 4, such that the temperature sensor **130B** or a circuit board supporting the temperature sensor **130B**, is not exposed to the airflow **110**, such as through an opening or “window” in the airflow housing **140**. As a result, the temperature sensed by the temperature sensor **130B** is not driven by the airflow **110**. Rather, the temperature sensed by the temperature sensor **130B** is substantially driven by the temperature of the surrounding air that is in direct contact with the temperature sensor **130B**, as well as a current driven through temperature sensor **130B** by the heating voltage during the heating of the airflow tube **104** and/or temperature sensing operations, for example.

[0064] In one embodiment, the temperature sensor **130B** is located within the airflow pathway **128** by attaching the temperature sensor **130B** to the interior wall **142** of the airflow housing **140** or by supporting the temperature sensor **130B** within a thermally conductive structure **150** that is embedded in the interior wall **142** or protrudes into the airflow pathway **128** from the interior wall **142**, as shown in FIG. 5, for example. Accordingly, the temperature of the temperature sensor **130B** supported within the airflow pathway **128** is substantially driven by the airflow **110**.

[0065] FIGS. 6 and 7 are simplified diagrams of airflow tubes **104A** and **104B** respectively having three and four electrical signal conductors, in accordance with embodiments of the present disclosure. Each of the airflow tubes **104A** and **104B** include the temperature sensor **130** on the machine side **132** of the flexible tube **112** as either a component of the machine connector **114** (temperature sensor **130B**) or as a component of the flexible tube **112** (temperature sensor **130A**). In some embodiments, the machine connector **114** includes an electrical connector **152** that connects to a corresponding connector of the airflow generator machine **102**. The electrical connector **152** includes electrical contacts **154** that are each connected to an electrical signal conductor of the airflow tube **104**, through which electrical signals, such as those generated by the airflow generator machine **102**, are conducted.

[0066] The electrical connector **152** of the airflow tube **104A** includes an electrical contact **154A** that is connected to the heater wire **122** through a conductor **156**, an electrical contact **154B** that is connected to the heater wire **124** through a conductor **158**, and an electrical contact **154C** that is connected to the temperature sensor **130** through a conductor or sensor wire **160**. In one embodiment, the temperature sensor **130** is connected in parallel between the electrical contact **154B** and the electrical contact **154C** using the sensor wire **160**. In one embodiment, a self-resetting electrical fuse **161** may be placed in series with the heater wires **122** and **124** to open the circuit in the event of a short circuit, current spike (e.g., around 1.5-1.75 amps) or voltage spike through the heater wire **122** or **124**, and self-reset when the fault condition is removed.

[0067] As mentioned above, the temperature sensor **130** may comprise a thermistor having a resistance that varies with its temperature. Thus, the temperature sensor **130** may have a “temperature signal” corresponding to its resistance, which may be determined based on a voltage V_s between the electrical contacts **154B** and **154C**, when a sense current $I_{sub.s}$ conducted through the temperature sensor **130** and the electrical contacts **154B** and **154C**. A scalar resistance **162** (FIGS. 6 and 7) may be placed in series with the temperature sensor **130** to tune the temperature

signal, if necessary, such that a desired voltage range for Vs is achieved in response to the sense current I.sub.s over a range of ambient temperatures.

[0068] During a heating operation, a heating current I.sub.h (e.g., PWM current) is conducted through the heater wires **122** and **124** and the electrical contacts **154A** and **154B**. The temperature signal (Vs) of the airflow tube **104** is typically sensed or measured by the airflow generator machine **102** during a period when the heating current I.sub.h is zero, in accordance with conventional techniques.

[0069] The electrical connector **152** of the airflow tube **104B** may be formed similarly to the electrical connector of the airflow tube **104**. For example, the electrical connector **152** of the airflow tube **104B** may include the electrical contacts **154A** and **154B** that are respectively connected to the heater wires **122** and **124** through conductors **156** and **158**. The electrical connector **152** of the airflow tube **104B** also includes electrical contacts **154C** and **154D** that are dedicated to the temperature sensor **130**. That is, the temperature sensor **130** and the scaler resistance **162** (if present) are connected in series with the electrical contacts **154C** and **154D** through conductors or sensor wires **160** and **164**. This configuration allows for continuous or periodic monitoring of the temperature signal corresponding to the voltage Vs between the electrical contacts **154C** and **154D** using the sense current I.sub.s, even while the heating current I.sub.h is conducted through the heater wires **122** and **124** and the electrical contacts **154A** and **154B**.

[0070] Accordingly, locating the temperature sensor **130** on the machine side **132** of the airflow tube **104** allows the one or more sensor wires (**160** and **164**) that are connected to the temperature sensor **130** to be completely eliminated from the flexible tube unlike the sensor wires (**232**, **234** and **236**) of the airflow tubes **208** of the prior art, or at least formed shorter than the sensor wires in the flexible tube of the airflow tubes **208** (FIGS. **12** and **13**). For example, when the electrical connector **152** includes the temperature sensor **130**, the only electrical signal conductors or wires included in the flexible tube **112** may be the heater wires **122** and **124**. Thus, in some embodiments, the flexible tube **112** consist of only two electrical signal conductors in the form of the heater wires **122** and **124**, rather than the three or more electrical signal conductors of the flexible tubes of conventional airflow tubes that extend along the entire length of the flexible tube. As a result, the airflow tube **104** includes significant advantages over conventional airflow tubes.

[0071] For example, the airflow tube **104** according to embodiments of the present disclosure has a significantly reduce complexity relative to conventional airflow tubes having flexible tubes with three or more electrical signal conductors extending their entire length resulting in reduced manufacturing costs and lower cost to the end user, while also eliminating a potential source of malfunction, such as a breakage of one of the sensor wires due to bending or crimping of the flexible tube. Additionally, the reduced number of electrical signal conductors and length of the electrical signal conductors within the flexible tube **112** reduces the weight of the airflow tube **104**, improves its flexibility over its entire length or over the patient interface side **134**, thereby simplifying and improving use of the airflow tube **104**. Eliminating the sensor wires in the flexible tube **112** also increases patient safety by eliminating a source of electrical interference generated by the long sensor wires found in the airflow tubes **208** of the prior art, which can be hazardous to patients with pacemakers or other electronic implantable devices.

[0072] Furthermore, due to the location of the temperature sensor **130** on the machine side **132** of the airflow tube **104**, such as the temperature sensor **130B** on the machine connector **114** supported outside the airflow pathway **128**, which may be 3-4 feet or more from the patient interface end **120** for an airflow tube **104** having a length of 6 feet, the temperature sensed by the temperature sensor **130** is not affected by the patient's breath or the airflow **110**, which may not be the case in the conventional respiratory apparatus **200** having an airflow tube **208** in accordance with the prior art or at least a temperature sensor at the patient interface end **120**. As a result, instability issues with the sensed temperature due to the patient's breath are avoided by the airflow tube **104**, allowing for

more stable and accurate temperature measurements and simplified control of the heating of the flexible tube **112**.

[0073] In some embodiments, the three contact electrical connector **152** of the airflow tube **104A** (FIG. 6), the temperature sensor **130**, the conductors **156** and **158**, and the sensor wire **160** form an assembly **170**, such as shown in FIGS. 3 and 4. The assembly **170** simplifies manufacture of the airflow tube **104** by allowing an assembler to form the electrical circuit of the airflow tube **104** by connecting the conductor **156** to the heater wire **122** and connecting the conductor **158** to the heater wire **124** through a suitable conventional technique, such as soldering, for example, rather than having to connect each electrical contact **154** of the electrical connector **152** to its corresponding conductor when assembling the airflow tube **104**. A similar assembly may be formed with the electrical connector **152** having four electrical contacts **154A-D**, the conductors **156** and **158**, and the sensor wires **160** and **164** of the airflow tube **104B** (FIG. 7). Here the assembler must only connect the conductors **156** and **158** to the corresponding heater wires **122** and **124** to form the desired electrical circuit of the airflow tube **104**.

[0074] In one embodiment, the airflow housing **140** includes an assembly connector **172** that allows an assembler to secure the assembly **170** to the airflow housing **140** by hand, as shown in FIGS. 3 and 4, to improve the efficiency at which the airflow tube **104** may be manufactured. For example, the assembly connector **172** may be located on the exterior of the airflow housing **140** and include tabs **174** or another suitable structure, in which a portion of the assembly **170**, such as the electrical connector **152**, may be pressed into by hand to secure the assembly **170** to the airflow housing **140**.

[0075] FIG. 8 is a simplified side view of a respiratory apparatus **100**, which includes an airflow generator machine **102** and the airflow generator airflow tube **104** formed in accordance with one or more embodiments described herein. In some embodiments, the airflow generator machine **102** includes an airflow generator **180** that is controlled by an airflow generator controller **182** to produce the desired airflow **110**, in accordance with conventional respiratory apparatuses. Thus, the airflow generator **180** may comprise a conventional airflow generator (e.g., blower fan), such as that used in CPAP units, for example. The airflow generator **180** may also include a conventional humidifier for adding water vapor to the airflow **110** in response to control signals from the airflow generator controller **182**.

[0076] The machine connector **114** of the airflow tube **104** connects to the airflow generator machine **180** at the output port **126**, which directs the airflow **110** into the airflow pathway **128**. The airflow **110** is delivered through the airflow pathway **128** of the machine connector **114**, the flexible tube **112** and the patient interface connector **118**, and is delivered to an attached patient interface **106**.

[0077] The respiratory apparatus **100** also includes a heating unit **184** that is controlled by a heater controller **186** to drive the heating current $I_{\text{sub.h}}$ through the heater wires **122** and **124** to heat the flexible tube **112** and the airflow **110** using the temperature sensor **130** that is located on the machine side **132** of the airflow tube **104**. The temperature sensor **130** of the airflow tube **104** may take the form of a temperature sensor **130B-1** (solid lines) that is attached (e.g., directly attached) to the machine connector **114** and located outside the airflow pathway **128**, such as temperature sensor **130B** shown in FIG. 3, a temperature sensor **130B-2** that is attached (e.g., directly attached) to the machine connector **114** and contained in the airflow pathway **128**, a temperature sensor **130A-1** that is attached (e.g., directly attached) to the flexible tube **112** and located outside the airflow pathway, or a temperature sensor **130A-2** that is attached (e.g., directly attached) to the flexible tube **112** and located within the airflow pathway **128**.

[0078] In one embodiment, the heater controller **186** is controlled based on a temperature sensed by the temperature sensor **130B** that is attached to or supported by the machine connector **114** outside the airflow pathway **128** and without using a temperature sensor that is configured to substantially directly measure the temperature of the airflow **110**, such as using a temperature sensor that is

supported in the airflow pathway or a temperature sensor that is exposed to the airflow **110** traveling through the airflow pathway **128**, for example.

[0079] In some embodiments, the temperature sensor **130** may be replaced by a temperature sensor **183** that is upstream from the machine connector **114** and the airflow tube **104** relative to the airflow **110**. For example, a temperature sensor **183A** may be located at the output port **126** of the airflow generator **180** or a temperature sensor **183B** may be located at an input port **185** of the airflow generator **180** through which the airflow is drawn. The temperature sensors **183A** and **183B** may be positioned either outside the airflow pathway through which the airflow **110** travels through the airflow generator **180** or within the airflow pathway through which the airflow **110** travels through the airflow generator **180**. A temperature signal from the temperature sensor **183** may be used by the heater controller **186** to control the heating unit **184** in the same manner as the temperature signal V_s is used.

[0080] The heating unit **184** and the heater controller **186** may be connected to the heater wires **122** and **124**, and the temperature sensor **130** through any of the configurations described herein. For example, the airflow generator machine **102** may include a machine electrical connector **187** that cooperates with the electrical connector **152** of the machine connector **114** to connect the heating unit **184** and the heater controller **186** to the contacts **154** of the electrical connector **152** and form an electrical circuit through which the heating current $I_{sub.h}$ may be delivered through the heater wires **122** and **124** by the heating unit **184**, and the temperature of the temperature sensor **130** may be sensed by the heater controller **186**.

[0081] The airflow tube **104** shown in FIG. **8** includes the electrical connector **152** of the airflow tube **104A** of FIG. **6** having three electrical contacts **154A-C**. However, the airflow tube **104** may take the form of the airflow tube **104B** of FIG. **7** having an electrical connector **152** that includes **4** electrical contacts **154A-D**, or the airflow tube **104** may have another suitable electrical connector **152**.

[0082] The heater controller **186** may sense the temperature of the temperature sensor **130** based on a temperature signal corresponding to a voltage (V_s) across the temperature sensor. The heater controller **186** may include conventional circuitry for processing the temperature signal (V_s) and comparing a value indicated by the temperature signal to values corresponding to the upper and lower limits of a set-point temperature range, such as that disclosed in U.S. Pat. No. 8,733,349, for example.

[0083] The heater controller **186** may be programmed based on empirical studies to control the heating unit **184** to deliver the proper amount of electrical power (heater current $I_{sub.h}$) through the heater wires **122** and **124** based on the temperature indicated by the temperature signal V_s . That is, based on the set-point temperature range and the temperature indicated by the temperature signal, the heater controller **186** may control the heating unit **184** to deliver the electrical current $I_{sub.h}$ (PWM current) through the heater wires **122** and **124** to heat the flexible tube **112** and the airflow **110** and/or prevent a rainout condition in the flexible tube **112**.

[0084] FIG. **9** is a flowchart of an example of a method of controlling the heating of the airflow tube **104** using the heater controller **186**, in accordance with embodiments of the present disclosure. When, for example, the airflow generator **180** of the respiratory apparatus **100** is activated, it produces the airflow **110** and the heater controller **186** detects a temperature of the temperature sensor **130** based on the temperature signal V_s , as indicated at **188**. At **189**, the heater controller **186** compares a value indicated by the temperature signal V_s , possibly after conventional processing by the electronics of the heater controller **186**, to a set-point temperature range (e.g., upper and/or lower value limits). The set-point temperature range may be set by the user or determined and preset by medical personnel. The upper value limit of the set-point temperature range may be set to meet the appropriate safety requirements of the FDA. The separation between the lower and upper value limits of the range of set-point values is set to be sufficiently wide to prevent switching between heating and non-heating modes too often and too fast.

[0085] If the value indicated by the temperature signal is within the set-point temperature range, the method continues to monitor the temperature signal Vs by returning to step **188**, such as after a predetermined delay.

[0086] If the value indicated by the temperature signal Vs is not within the set-point temperature range, the heater controller **186** determines whether the value indicates a temperature that is above or below the set-point temperature range at **190**.

[0087] If the value indicated by the temperature signal Vs is higher than the set-point temperature range (e.g., upper value limit), the heater controller **186** may deactivate the heating current I.sub.h, such as by terminating power to the heating unit **184**, as indicated at **191**.

[0088] If the value indicated by the temperature signal Vs is lower than the set-point temperature range (e.g., lower value limit), the heater controller **186** controls the heating unit to deliver a current I.sub.h through the heater wires **122** and **124**, as indicated at **192**. The electrical power delivered by the current I.sub.h may be increased or decreased based on a difference between the value indicated by the temperature signal Vs and the setpoint temperature range, in accordance with conventional techniques.

[0089] When the heating unit **184** is powered by direct current (DC) voltage, the heater controller **186** may adjust the current delivered through the heater wires **122** and **124** using

[0090] Analog or pulse width modulation (PWM) control of the DC voltage and sense the temperature by sampling the temperature signal Vs (step **188**) during a PWM off cycle. If an alternating current (AC) is used, the frequency and voltage level of the AC current through the heater wires **122** and **124** can be adjusted to provide the desired heating effect. As in any of the aforementioned cases, the temperature could be read in step **188** continuously or intermittently as needed to obtain the best temperature control requirement.

[0091] Following step **191** or **192**, the method returns to step **188** and continues as described above, until the respiratory apparatus **100** is deactivated. A delay may be imposed before performing step **188** in order to allow the temperature to stabilize.

[0092] In some embodiments, the respiratory apparatus **100** may be configured to automatically set heater control parameters of the heater controller **186** and/or the airflow generator controller **182** based on a type of airflow tube **104** that is connected to the airflow generator machine **102**, which may be identified based on a resistance of the temperature sensor **130** detected using the heater controller **186**. As an example, a 15 mm internal diameter airflow tube **104** may include a temperature sensor **130** with a thermistor value of 10 k Ω and a 12 mm internal diameter heated tube may include a temperature sensor **130** with a thermistor value of 2 k Ω . Thus, the heater controller **186** and/or the airflow generator controller **182** may discriminate between different resistance values corresponding to the temperature sensor **130** that would indicate the diameter (e.g., 19 mm, 15 mm or 12 mm, etc.) and/or length of the airflow tube **104** in use to permit changes in the operation of the heater controller **186** and/or airflow generator controller **182** based on the type and characteristics of airflow tube **104** that is being used.

[0093] Thus, any compensation for air path performance (e.g., heating and temperature change of the airflow **110** from the machine end **116** to the patient interface end **120**) can be adjusted automatically (without user intervention) for each type of airflow tube **104** by the heater controller **186**, if required, such as electronically or through a look-up table based on the resistance of the temperature sensor **130** of the attached airflow tube **104**. It should be appreciated that more than two types of tubes **104** may be detected by the airflow generator machine **102** using multiple comparator values, gains and thermistor styles. Detection of airflow tube types can also be used to allow amplifier gain designs provided by the electronics of the heater controller **186** to increase the amplitude of the temperature sense signals for a lower sensitivity (higher value NTC thermistor) circuit. This could also allow for adjustment in the airflow tube design of the signal gain so that the same tube may be used for different tube type requirements (i.e. different internal diameters).

[0094] The airflow generator controller **182** and the heater controller **186** may take on any suitable

form, including the form of a single controller that performs their functions. A simplified diagram of an example of a controller **193** that may form the controller **182**, the heater controller **186** or a combined controller, is shown in FIG. **10**. The controller **193** may include one or more processors **194** and memory **195**. The one or more processors **194** are configured to perform various functions (e.g., control of the airflow generator **180**, control of the heating unit **184**, etc.), in response to the execution of instructions contained in the memory **195**.

[0095] The one or more processors **194** may be components of one or more computer-based systems, and may include one or more control circuits, microprocessor-based engine control systems, and/or one or more programmable hardware components, such as a field programmable gate array (FPGA). The memory **195** represents local and/or remote memory or computer-readable media. Such memory **195** comprises any suitable patent subject matter eligible computer readable media and does not include transitory waves or signals. Examples of the memory **195** include conventional data storage devices, such as hard disks, CD-ROMs, optical storage devices, magnetic storage devices and/or other suitable data storage devices. The controller **193** may include circuitry **196** for use by the one or more processors **194** to receive input signals **197** (e.g., airflow generator control signals, temperature signals, settings, etc.), issue control signals **198** (e.g., control signals for the airflow generator **180**, the heating unit **184**, etc.) and/or communicate data **199**, such as in response to the execution of the instructions stored in the memory **195** by the one or more processors **194**.

[0096] Although the embodiments of the present disclosure have been described with reference to preferred embodiments, workers skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the present disclosure.

[0097] It will further be understood that any reference herein to known prior art does not, unless the contrary indication appears, constitute an admission that such prior art is commonly known by those skilled in the art to which the invention relates.

[0098] Specific details are given in the above-description to provide a thorough understanding of the embodiments. However, it is understood by those of ordinary skill in the art that the embodiments may be practiced without these specific details. For example, circuits, systems, networks, processes, frames, supports, connectors, motors, processors, and other components may not be shown, or may be shown in block diagram form in order to not obscure the embodiments in unnecessary detail.

[0099] As used herein, when one or more functions or process steps are described as being performed by a controller (e.g., a specific controller), one or more controllers, at least one controller, a processor (e.g., such as a specific processor), one or more processors or at least one processor, embodiments include the performance of the function(s) by a single controller or processor, or multiple controllers or processors, unless otherwise specified. Furthermore, as used herein, when multiple functions are performed by at least one controller or processor, all of the functions may be performed by a single controller or processor, or some functions may be performed by one controller or one processor, and other functions may be performed by another controller or processor. Thus, the performance of one or more functions by at least one controller or processor does not require that all of the functions are performed by each of the controllers or processors.

Claims

1. A respiratory apparatus airflow tube configured to deliver an airflow from an airflow generator machine to a patient interface, the airflow tube comprising: a flexible tube including a machine end, a patient end, first and second heater wires extending from the machine end toward the patient end, and an airflow pathway extending within the flexible tube from the machine end to the patient end; a patient interface connector attached to the patient end and configured to attach to a patient

interface; and a machine connector attached to the machine end and configured to attach to an airflow generator machine, the machine connector including: an airflow housing having an interior wall defining a portion of the airflow pathway; and a temperature sensor located outside the airflow pathway, wherein an airflow generated by the airflow generator machine travels through the airflow pathway including the airflow housing.

2. The airflow tube according to claim 1, wherein the machine connector includes an airflow tube electrical connector configured to connect to a machine electrical connector of the airflow generator machine, the airflow tube electrical connector comprising: a first electrical contact connected to the first heater wire; a second electrical contact connected to the second heater wire; and a third electrical contact connected to the temperature sensor.

3. The airflow tube according to claim 2, wherein the temperature sensor is connected across the second and third electrical contacts.

4. The airflow tube according to claim 2, wherein: the airflow tube electrical connector comprises a fourth electrical contact connected to the temperature sensor; and the temperature sensor is connected in series with the third and fourth electrical contacts.

5. The airflow tube according to claim 2, wherein the machine connector includes an electrical fuse in series with the first or second heater wire.

6. The airflow tube according to claim 1, wherein the temperature sensor is on an exterior side of the airflow housing.

7. The airflow tube according to claim 1, including an assembly attached to the airflow housing comprising: the temperature sensor; a first conductor connected to the first heater wire; a second conductor connected to the second heater wire; a third conductor connected to the temperature sensor; and an airflow tube electrical connector configured to connect to a machine electrical connector of the airflow generator machine comprising: a first electrical contact connected to the first conductor; a second electrical contact connected to the second conductor; and a third electrical contact connected to the third conductor.

8. The airflow tube according to claim 7, wherein the airflow housing includes an assembly connector that secures the electrical connector to the airflow housing.

9. A respiratory apparatus comprising: an airflow generator machine comprising: an airflow generator configured to drive an airflow through an output port; a heating unit; and a heater controller; and an airflow tube comprising: a flexible tube having a machine end, a patient end, and first and second heater wires extending from the machine end toward the patient end, and an airflow pathway extending within the flexible tube from the machine end to the patient end; a patient interface connector attached to the patient end and configured to attach to a patient interface; and a machine connector attached to the machine end of the flexible tube and connected to the output port, the machine connector including: an airflow housing having an interior wall defining a portion of the airflow pathway; and a temperature sensor located outside the airflow pathway, wherein: an airflow generated by the airflow generator machine travels through the airflow pathway including the airflow housing; and the heater controller is configured to control the heating unit to drive a current through the first and second heater wires using the temperature sensor.

10. The apparatus according to claim 9, wherein: the airflow machine includes a machine electrical connector; the airflow tube includes an airflow tube electrical connector configured to connect to the machine electrical connector, the airflow tube electrical connector comprising: a first electrical contact connected to the first heater wire; a second electrical contact connected to the second heater wire; and a third electrical contact connected to the temperature sensor.

11. The apparatus according to claim-9 10, wherein the temperature sensor is connected across the second and third electrical contacts.

12. The apparatus according to claim 11, wherein the heater controller controls the current through the first and second heater wires based on a voltage between the second and third electrical

contacts, which varies based on a temperature of the temperature sensor.

13. The apparatus according to claim 10, wherein: the airflow tube electrical connector comprises a fourth electrical contact connected to the temperature sensor; the temperature sensor is connected in series with the third and fourth electrical contacts; and the heater controller controls the current through the first and second heater wires based on a voltage between the third and fourth electrical contacts, which varies based on a temperature of the temperature sensor.

14. The apparatus according to claim 11, wherein the temperature sensor is located on an exterior side of the airflow housing.

15. The apparatus according to claim 11, wherein the airflow tube includes an assembly attached to the airflow housing comprising: the temperature sensor; a first conductor connected to the first heater wire; a second conductor connected to the second heater wire; a third conductor connected to the temperature sensor; and an airflow tube electrical connector connected to the machine electrical connector, the airflow tube electrical connector comprising: a first electrical contact connected to the first conductor; a second electrical contact connected to the second conductor; and a third electrical contact connected to the third conductor.

16. The apparatus according to claim 15, wherein the airflow housing includes an assembly connector that secures the electrical connector to the airflow housing.

17. The apparatus according to claim 9, including a patient interface comprising a mask attached to the patient interface connector.

18. The apparatus according to claim 10, wherein the machine connector includes an electrical fuse in series with the first or second heater wire.

19. A respiratory apparatus airflow tube configured to deliver an airflow from an airflow generator machine to a patient interface, the airflow tube comprising: a flexible tube including a machine end, a patient end, first and second heater wires extending from the machine end toward the patient end, and an airflow pathway extending within the flexible tube from the machine end to the patient end; a patient interface connector attached to the patient end of the flexible tube and configured to attach to a patient interface; and a machine connector attached to the machine end of the flexible tube and configured to attach to an airflow generator machine, the machine connector including: an airflow housing having an interior wall defining a portion of the airflow pathway; a temperature sensor; an electrical connector comprising: a first heater wire contact; a second heater wire contact; a first temperature sensor contact; and a second temperature sensor contact; a first conductor connecting the first heater wire contact to the first heater wire; a second conductor connecting the second heater wire contact to the second heater wire; and third and fourth conductors connecting the temperature sensor in series with the first and second temperature sensor contacts.

20. The respiratory apparatus according to claim 19, wherein: an airflow generated by the airflow generator machine travels through the airflow pathway including the airflow housing; and the temperature sensor is located outside the airflow pathway.

21. A method of controlling heating of an airflow tube of a respiratory apparatus, which includes: an airflow generator machine comprising: an airflow generator configured to drive an airflow through an output port; a heating unit; and a heater controller; and an airflow tube comprising: a flexible tube having a machine end, a patient end, first and second heater wires extending from the machine end toward the patient end, and an airflow pathway extending within the flexible tube from the machine end to the patient end; a patient interface connector attached to the patient end and configured to attach to a patient interface; and a machine connector attached to the machine end of the flexible tube and connected to the output port, the machine connector including: an airflow housing having an interior wall defining a portion of the airflow pathway; and a temperature sensor attached to the airflow housing, the method comprising: driving an airflow through the airflow pathway using the airflow generator; detecting a temperature using the temperature sensor at the machine connector; and controlling a current through the first and second heater wires using the heater controller based on a temperature consisting of the detected

temperature, wherein controlling the current is not affected by a breath of a patient received at the patient interface end.

22. The method according to claim 21, wherein the airflow tube does not include a temperature sensor at the patient interface connector.

23. The method according to claim 22, wherein the temperature sensor is located within the airflow pathway.

24. The method according to claim 22, wherein the temperature sensor is located outside the airflow pathway.

25. A respiratory apparatus airflow tube configured to deliver an airflow from an airflow generator machine to a patient interface, the airflow tube comprising: a flexible tube including a machine end, a patient end, first and second heater wires extending from the machine end toward the patient end, and an airflow pathway extending within the flexible tube from the machine end to the patient end; a patient interface connector attached to the patient end and configured to attach to a patient interface; and a machine connector attached to the machine end and configured to attach to an airflow generator machine, the machine connector including: an airflow housing having an interior wall defining a portion of the airflow pathway; a temperature sensor; a scaler resistance connected in series with the temperature sensor; and an electrical connector comprising: a first electrical contact connected to the first heater wire; a second electrical contact connected to the second heater wire; and a third electrical contact connected to the temperature sensor.

26. The airflow tube according to claim 25, wherein the temperature sensor is connected across the second and third electrical contacts.

27. The airflow tube according to claim 25, wherein: the airflow tube electrical connector comprises a fourth electrical contact; and the temperature sensor and the scaler resistance are connected in series with the third and fourth electrical contacts.

28. A respiratory apparatus airflow tube configured to deliver an airflow from an airflow generator machine to a patient interface, the airflow tube comprising: a flexible tube including a machine end, a patient end, first and second heater wires extending from the machine end toward the patient end, and an airflow pathway extending within the flexible tube from the machine end to the patient end; a patient interface connector attached to the patient end and configured to attach to a patient interface; and a machine connector attached to the machine end and configured to attach to an airflow generator machine, the machine connector including: an airflow housing having an interior wall defining a portion of the airflow pathway; a temperature sensor; and an electrical connector comprising: a first electrical contact connected to the first heater wire; a second electrical contact connected to the second heater wire; a third electrical contact; and a fourth electrical contact, wherein the temperature sensor is connected in series with the third and fourth electrical contacts.
